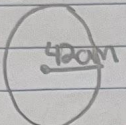


HWK #4

①  $\theta?$
 arc length 21cm & 5.5m

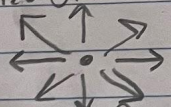
Ⓐ $\theta = \frac{l}{r} \rightarrow \frac{21 \times 10^{-2} \text{ m}}{42 \times 10^{-2} \text{ m}} = \boxed{0.5 \text{ rad}}$

Ⓑ $\theta = \frac{5.5 \text{ m}}{42 \times 10^{-2} \text{ m}} = \boxed{13.095 \text{ rad}}$

②  $\theta?$
 15m² & 25cm²

Ⓐ $\Omega = \frac{A}{r^2 \text{ sr}} \rightarrow \frac{15 \text{ m}^2}{(750 \times 10^{-3})^2 \text{ m}} = \boxed{26.67 \text{ sr}}$

Ⓑ $\Omega = \frac{25 \times 10^{-2} \text{ m}}{(750 \times 10^{-3})^2 \text{ m}} = \boxed{0.45 \text{ sr}}$

③ 18W = Power
 12cm = r

 5mm² = Area
 25cm = distance

radiant power?
 radiant intensity?
 radiant emittance?
 radiance?
 irradiance?

Ⓐ Radiant power (Φ) = $\boxed{18 \text{ W}}$

Ⓑ Radiant Intensity = $\frac{\Phi}{4\pi \text{ sr}} \rightarrow \frac{18 \text{ W}}{4\pi} = \boxed{1.43 \text{ W/sr}}$

③ Radiant Emittance $\rightarrow \frac{P}{4\pi r^2}$

$\rightarrow \frac{18W}{4\pi(1.2 \cdot 10^{-2}m)^2} = \boxed{9947.18 W/cm^2}$

④ Irradiance $\rightarrow \frac{18W}{4\pi(5 \cdot 10^{-3})^2} \times 25 \cdot 10^{-2} = \boxed{14,323.94 W/m^2}$

⑤ Radiance $\rightarrow \frac{18W}{4\pi(4\pi)(1.2 \cdot 10^{-2})^2} = \boxed{791.57 W/m^2}$

⑥ $P = 25mW$ } radiant?
intensity?

$\frac{P}{4\pi sr} = \frac{25 \cdot 10^{-3}W}{4\pi} = \boxed{1.989 mW/sr}$

⑦ $O_3 @ 4727^\circ C \} \lambda_{max}$

$\lambda_{max} = \frac{2898 \mu m \cdot K}{K}$

① $4727^\circ C + 273.15 = 5000.15^\circ K$

② $\frac{2898 \cdot 10^{-6}m}{5000.15^\circ K} = \boxed{579.58 nm}$

⑧ ① $F - 32 \times 5/9 \rightarrow (734 - 32) 5/9 = 390^\circ C$

② $390^\circ C + 273.15 = 663.15^\circ K$

③ $\frac{2898 \cdot 10^{-6}m}{663.15^\circ K} = \boxed{4.37 \mu m}$

⑦ Black body @ 5500K } 0.5 - 2.5 μm
spectral radiance

$$L_\lambda = 2\pi \cdot \frac{hc^2}{\lambda^5} \cdot \frac{1}{e^{hc/\lambda kT} - 1}$$

$$\textcircled{a} \rightarrow 2\pi \cdot \frac{(6.626 \times 10^{-34})(3 \cdot 10^8)^2}{(0.5 \times 10^{-6})^5} \cdot \frac{1}{e^{\frac{(6.626 \cdot 10^{-34})(3 \cdot 10^8)}{(0.5)(5500)(1.38 \cdot 10^{-23})}} - 1}$$

$$= \boxed{2.73 \cdot 10^{14} \text{ W/m}^2 \cdot \text{sr}}$$

$$\textcircled{b} 2\pi \cdot \frac{(6.626 \times 10^{-34})(3 \cdot 10^8)^2}{(2.5 \cdot 10^{-6})^5} \cdot \frac{1}{e^{\frac{(6.626 \cdot 10^{-34})(3 \cdot 10^8)}{(2.5)(5500)(1.38 \cdot 10^{-23})}} - 1}$$

$$= \boxed{3.66 \cdot 10^{12} \text{ W/m}^2 \cdot \text{sr}}$$

